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Dynamic quantification anti-fraud machine learning model for real-time transaction fraud detection in banking

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Abstract
As the scenario of telecom network fraud intensifies, the development of anti-fraud models has become a focal point in financial technology, including banking sectors. Traditional anti-telecom fraud models, which primarily rely on expert rules and machine learning methodologies, are beset with limitations such as poor real-time performance and high false favourable rates, leading to biased predictive outcomes. This paper introduces a novel anti-fraud model: The Dynamic Quantification Anti-Fraud Model grounded in Real-time Transaction Flows to address these issues. This model combines multiple F-Beta features linearly, employs a DML (Double Machine Learning) framework for feature weight quantification, and conducts instantaneous risk assessments of account transactions. By dynamically adjusting feature weights based on historical single-feature false favourable rates and the branch's anti-fraud tolerance capacity, it significantly improves model performance and the efficiency of fraud prevention. Using transaction detail data of implicated cards from a case bank spanning January to June 2023 as an example for modelling, the validation sample recall rate reached 0.506. Post-deployment, the model facilitated a 30-percentage point reduction in the proportion of implicated cards within six months at the bank, alongside a substantial drop in its ranking among peers concerning the number of fraud cases. This comprehensive improvement solidly validates the precision and efficacy of the proposed model.

Keywords Anti-fraud model · Dynamic quantification · Implicated card prediction · Real-time transaction streams · Double machine learning

1 Introduction

Bank accounts, as the fundamental platform for fund flows, facilitate economic transactions but also inadvertently serve as tools for illegal activities by criminals [1]. With the flourishing development of the Internet economy, the situation regarding telecom network fraud has become increasingly severe [2]. Strengthening supervision over bank accounts has emerged as a potent weapon against such criminal activities, given their role as a necessary conduit for illicit fund transfers [3]. Effective account management systems enable banks to prevent illegal activities during account opening, continuously monitor transactions, and promptly intervene during account usage [4]. Advanced account monitoring systems allow banks to identify and flag suspicious transactions [5], promptly issue alerts [6], and disrupt or prevent illegal transactions, safeguarding public assets, maintaining a sound market economic order, and supporting economic

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health. Internet banking has increased rapidly during the past decade, and services like e-commerce, online payment systems, working from home, online banking, and social networking have also been widely introduced [7].

Recently, research on telecom fraud detection methods based on bank transaction records has been limited and divided into two categories: conventional expert rule-based methods and machine learning algorithms [8–10]. Expert rule-based methods, relying on multi-disciplinary expertise within banks [11], establish rules to identify abnormal account behaviour and assess fraud risk [12]. These rules are computationally swift, taking only 1–2 s per rule, enabling the freezing of accounts before fraudsters can transfer funds. Nonetheless, generalization capabilities are lacking in adapting to dynamic and complex fraud scenarios. Overreliance on subjective judgments by experts [13] and the singularity of rule dimensions increase misidentification rates, escalating operational costs and customer complaints for banks. For example, Islam et al. [14] developed rules to detect fraud transactions with the rule-based model using two benchmark datasets. It was shown that the proposed rule-based model reached an accuracy and precision of 0.99 compared to several existing models. Moreover, Yang et al. [15] proposed an innovative ensemble BRB (belief rule base) model to solve the credit card fraud detection problem, which achieved excellent performance on fraud detection and effectively mitigated the impact of an imbalanced dataset. Compliance and risk management services employed to identify online fraud have shown much interest in AI and machine learning models [16, 17].

Fraudsters are more active in their attacks on credit card transactions than ever before. With the advancement in data science and machine learning, various algorithms have been developed to determine whether a transaction is fraudulent [18–21]. Some machine learning algorithms, such as Random Forests [22, 23], XGBoost [24], and LGBM [25], analyze multi-dimensional client features through regression to predict the likelihood of an account being involved in fraud. These algorithms possess strong predictive power, enhancing the accuracy of telecom fraud detection to a certain extent. It is necessary because credit card fraud detection is a classification and prediction problem. For instance, Afriyie et al. [26] investigated the supervised machine learning algorithm for detecting and predicting fraud in credit card transactions. It was shown that random forest provided a maximum accuracy of 96% and was recommended as the most appropriate machine learning algorithm. However, due to their computationally intensive nature and high resource demands [27], they struggle to cope with the vast volume of transaction data typical in banking. The latency introduced by lengthy training periods hampers proactive fraud prevention. Machine learning algorithms are used to classify and predict financial transactions as either fraudulent or not fraudulent [28–30].

This paper proposes a novel Dynamic Quantification Anti-Fraud Model based on real-time transaction flows. Its core strengths include integrating multi-dimensional expert rule features to quantify risk indicators for bank accounts, ensuring model accuracy, rapidly generating analysis results based on real-time data to handle mass transaction data instantly, and emphasizing preemptive warnings by deeply analyzing historical transaction patterns, frequency, amount distributions, time modes, and inter-account interactions, to promptly detect abnormal behaviour resembling past fraud cases, effectively halting illicit fund transfers and safeguarding public property.

2 Real-time anti-fraud dynamic quantification system model

The ultimate objective of establishing a data model is to apply real-time data in practical business scenarios, ensuring its implementation across diverse projects. The study utilizes a dataset based on all disconnection card clues issued by the People's Bank of China received by the Agricultural Bank of China, Zhejiang Branch, from January 2022 to June 2023. Positive samples are accounts with an annual average financial asset below 5000 yuan and at least one instance of a transaction exceeding 3000 yuan from an unfamiliar counterparty. Complete transaction histories preceding any fraudulent activities of these accounts are analyzed to identify potential risk indicators. A control group of non-implicated normal accounts is selected following identical criteria to ensure comparability. The paper implies a balanced dataset (equal criteria for positive/negative samples), which is unrealistic for fraud detection where fraud is rare (e.g., $< 0.1\%$ of transactions in real-world data). Balancing classes may improve model training but risks overfitting to synthetic data. In practice, fraud detection models often use imbalanced datasets (e.g., 99.9% normal, 0.1% fraud) and employ techniques like SMOTE or cost-sensitive learning to address bias [31].

This model is specifically designed for real-time transaction monitoring scenarios, with a core function of conducting immediate risk assessments on specific anomaly transactions. Upon detecting a real-time transaction involving a single transfer of over 3,000 yuan from an unknown party, the system promptly connects to a dedicated interface provided by the head office, retrieving the latest transaction data related to the account and fetching the complete transaction records of the past month from the local database. It used a test set of 633 positive samples (implicated cards) in July 2023 but did not

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disclose the size of the training or validation datasets. 633 (test set) and an unspecified number in the training set (implicated cards with annual assets ≤ 5000 RMB and ≥ unknown party > 3000 RMB). Non-implicated normal accounts were selected using the same criteria (annual assets ≤ 5000 RMB and ≥ unknown party > 3000 RMB). The paper implies a balanced dataset (e.g., "control group of non-implicated normal accounts"). Normal accounts were chosen based on the same thresholds as positive samples: Annual average financial assets ≤ 5000 RMB. At least one single transaction > 3000 RMB to an unfamiliar counterparty. This selection is unusual because such transactions might inherently signal risk, potentially contaminating the "normal" group with borderline cases. Within a brief 5 to 10 s post-data collection, the dynamic quantification scoring model employs built-in algorithms to deeply analyze the transaction data, generating a risk assessment outcome for the transaction and determining whether to restrict the account's outgoing transaction capability. The process is illustrated in Fig. 1.

2.1 Weighted combination prediction model based on F_β

2.1.1 Definition of Basic risk indicator scores

Initially, we delve into the data of involved cards from one month prior within the training dataset and construct several distinct expert rules based on suspicious transaction behaviours. We calculate the F_β value for each feature according to the clues output by each expert rule. F_β (F-beta Score) is an evaluation metric used in machine learning and statistics to assess the performance of binary classifiers, particularly effective in scenarios requiring balancing different types of errors (false positives and false negatives) [32]. It is a weighted harmonic mean of Precision and Recall, defined by the formula:

$$F_\beta = \left(1+\beta^2\right)\frac{\text{Precision} \times \text{Recall}}{\beta^2\text{Precision} + \text{Recall}} \tag{1.1}$$

where:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{1.2}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{1.3}$$

Firstly, for all target bank accounts $\mathbf{X}$ in the 30-day transaction history leading up to the modelling point, we mark whether they involve suspected fraud features listed in Table 1, calculating the F_β for each such feature. Then, for each individual account x (where x belongs to $\mathbf{X}$), we combine all the involved suspects. F_β scores with weights to derive the risk assessment indicator for the account, constituting a multi-feature F_β weighted combination prediction model defined as:

$$F(x) = \sum_{i\in\mathbf{I}_x} w_i(F_\beta)_i, \mathbf{I}_x \subseteq I\# \tag{1.4}$$

here, $\mathbf{I}$ represents the complete set of all involved features, while $\mathbf{I}_x$ denotes the subset of suspected fraud features associated with account x. w_i is the weight assigned to feature i of the account, where i is part of $\mathbf{I}$. $(F_\beta)_i$ signifies the f-beta score of feature i for the account. According to expert experience, the starting value of β is set at 3, adjustable within the range $\{\beta : 2 \le \beta \le 3\}$. Choosing $\beta > 1$stems from the requirement for high recall in identifying involved accounts, accepting a certain level of false positives to ensure no potential fraud cases are overlooked. The higher the $F(x)$ coefficient, the greater the likelihood that the card is involved in fraud.

Furthermore, the description of each feature is briefly introduced. For example, the small amount test feature focuses on transactions of relatively low values. Financial institutions may monitor small-amount transactions because fraudsters sometimes use a series of small-value transactions to avoid detection. The abnormal transaction amount feature pertains to transactions with amounts that could be significantly larger or smaller than the card's regular transaction amounts. A

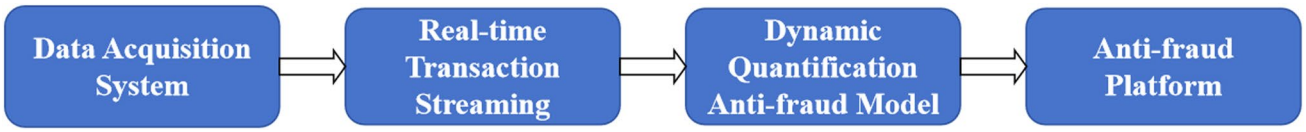


Fig. 1 Prediction procedure of dynamic quantification anti-fraud model based on real-time transaction streams

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Table 1 Data characteristics and F_β values for Implicated Cards in June 2023 at the Agricultural Bank of China, Zhejiang Branch ($\beta = 3$)

Data	Number of involved cards	Precision	Recall	F-beta
Small amount test 1	46	0.0981	0.0187	0.0203
Small amount test 2	41	0.0874	0.0201	0.0218
Small amount test 3	33	0.0704	0.0279	0.0297
Small amount test 4	8	0.0171	0.0231	0.0223
Abnormal transaction amount 1	7	0.0149	0.0029	0.0032
Abnormal transaction amount 2	6	0.0128	0.0038	0.0041
Abnormal transaction amount 3	4	0.0085	0.0003	0.0003
Abnormal transaction amount 4	1	0.0021	0.0046	0.0041
Frequent transfer 1	30	0.0640	0.0007	0.0007
Frequent transfer 2	29	0.0618	0.0011	0.0012
Unfamiliar counterparty transfer 1	190	0.4051	0.0024	0.0027
Unfamiliar counterparty transfer 2	81	0.1727	0.0013	0.0014
Unfamiliar counterparty transfer 3	43	0.0917	0.0026	0.0028
Unfamiliar counterparty transfer 4	67	0.1429	0.0068	0.0075
Abnormal transaction time	37	0.0789	0.0003	0.0003
Geographic anomaly	12	0.0256	0.0000	0.0000
Rapid consecutive transactions 1	94	0.2004	0.0001	0.0002
Rapid consecutive transactions 2	86	0.1834	0.0001	0.0001
Rapid consecutive transactions 3	66	0.1407	0.0001	0.0001
Rapid consecutive transactions 4	60	0.1279	0.0023	0.0025
Rapid consecutive transactions 5	40	0.0853	0.0013	0.0014

large-value transaction might indicate unauthorized access to the card, while a minimal amount could be part of a test by fraudsters or a sign of a miscalculation in a fraudulent scheme. The frequent transfer feature tracks the frequency of transfer activities associated with a card. High-frequency transfers, especially in a short period. It could be a sign of money-laundering, where funds are quickly moved around to obfuscate their origin or destination. Unfamiliar counterparty transfer feature is defined as sending money to unknown individuals or entities, which is a sign of fraud. The abnormal transaction time feature represents the unusual transaction hours.

Transactions outside these typical times may suggest that the card has been compromised, as an unauthorized user may not know the cardholder's normal usage schedule. The geographic anomaly feature focuses on the location where transactions take place. If a card is used in a location far from the cardholder's usual activity area, it could indicate fraud. The rapid consecutive transaction feature detects a series of transactions that occur quickly. Fraudsters may try to make as many transactions as possible in a short time before the cardholder or the bank realizes that the card has been compromised. Therefore, many rapid consecutive transactions can strongly indicate potential fraud. Table 1 lists 20 + expert-defined transactional features derived from historical fraud patterns, such as "Small amount test 1,""Unfamiliar counterparty transfer 1," and "Rapid consecutive transactions 1." These features are manually engineered based on suspicious behaviors observed in experimental data, including transaction amounts, frequency, time, and counterparty familiarity. Each feature acts as a binary classifier (e.g., "If Small amount test 1 is detected, classify as fraud"). The F-β scores in Table 1 quantify their performance: precision (fraud cases correctly identified by the rule), recall (proportion of actual fraud detected), and F-β (β = 3 emphasizes recall). For instance, "Small amount test 1" has an F-β of 0.0203 because it detects 1.87% of fraud cases (recall) but misclassifies 90.19% of non-fraud cases (low precision). The model combines these rules by weighting their F-β scores dynamically via t Double Machine Learning (DML) [33], ensuring high-risk features (e.g., "Unfamiliar counterparty transfer") dominate the final risk score while mitigating false positives through logarithmic transformation and regularization.

After establishing the definition of the base indicator scores, various algorithms can be employed to determine the weight of each feature. Traditional weighting methods such as the Analytic Hierarchy Process (AHP) [34] and Principal Component Analysis (PCA) [35] fail to address the collinearity among multiple involved features and the overfitting risks posed by strong assumptions. To tackle these issues, we adopt the most recent DML causal inference method to ascertain the weight of each feature [36]. Detailed hyperparameter tuning was explicitly shown for the LightGBM and XGBoost models used in the DML

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framework. It was suggested that parameters like num_leaves, learning_rate, and feature_fraction were likely optimized via grid search or Bayesian optimization to minimize residual loss. Early stopping was probably applied to prevent overfitting. The machine learning component employs the LightGBM algorithm to investigate the causal impact of a single involved feature $\mathbf{T}$, represented as $\mathbf{T} = \{t_x | t_x \in \mathbf{I}, x \in \mathbf{X}\}$, on the involved outcome $\mathbf{Y}$, denoted as $\mathbf{Y} = \{y_x | y_x = 0,1, x \in \mathbf{X}\}$, for the collective set of all involved accounts $\mathbf{X}$. It requires constructing two machine learning models, *f* and *g*, following the workflow outlined in Fig. 2, yielding the residuals as follows:

$$\mathcal{L}_g(Z) = -\sum_{x \in X} \left(g(z_x) - t_x\right) \tag{1.5}$$

$$\mathcal{L}_f(Z) = -\sum_{x \in X} \left(f(z_x) - y_x\right) \tag{1.6}$$

here, t_x represents a single feature variable, y_x denotes the involved feature variable, and $\mathbf{Z}$ symbolizes the set of all other features z_x, where $\mathbf{Z} = \{z_x | z_x \in \mathbf{I} \setminus \mathbf{T}, x \in \mathbf{X}\}$.

The residual $\mathcal{L}_g(\mathbf{Z})$ excludes the influence of other features $\mathbf{Z}$ on the single involved feature $\mathbf{T}$, and the residual $\mathcal{L}_f(\mathbf{Z})$ excludes the influence of other features $\mathbf{Z}$ on the involved feature $\mathbf{Y}$. Thus, by fitting $\mathcal{L}_g(\mathbf{Z})$ and $\mathcal{L}_f(\mathbf{Z})$ using XGBoost [37], we obtain the fitted causal relationship function $\theta\left(\mathcal{L}_g(\mathbf{Z})\right)$ between $\mathbf{T}$ and $\mathbf{Y}$ as follows:

$$\overset{\}}{\mathcal{L}}_f(\mathbf{Z}) = \theta\left(\mathcal{L}_g(\mathbf{Z})\right) = \sum_{k=1}^{K} f_k\left(\mathcal{L}_g(\mathbf{Z})\right), f_k \in F\# \tag{1.7}$$

where $\mathcal{F} = \left\{ I(\mathbf{x}) = w^*_{q(\mathbf{x})} \right\} \left(q : \mathbb{R}^m \to T^*, w^* \in \mathbb{R}^{T^*}\right)$ denotes the space of regression trees (is known as CART). Here, *q* represents the structure of each tree, mapping a sample to the corresponding leaf node index. T^* is the number of leaf nodes in the tree. Each I_k corresponds to an independent tree structure *q* and leaf node weight w^*. Unlike decision trees, each leaf node in every regression tree holds a continuous score represented by w^*_i for the *i*-th leaf node.

Ultimately, $\theta\left(\mathcal{L}_g(\mathbf{Z})\right)$ yields the causal effect of the single involved feature $t_x \in \mathbf{T}$ on the involved status feature $y_x \in \mathbf{Y}$, which can be incorporated into Eq. (1.4) as a weight, transformed as:

$$F(x) = \sum_{t_x \in \mathbf{I}_x} \theta_{t_x}\left(\mathcal{L}_g(\mathbf{Z})\right)(F_\beta)_{t_x}\# \tag{1.8}$$

here, $\mathbf{I}_x$ represents the set of suspected involved features for bank account *x*, $\theta_{t_x}\left(\mathcal{L}_g(\mathbf{Z})\right)$ indicates the fitting of feature t_x to y_x, serving as the weight of feature t_x. $(F_\beta)_{t_x}$ is the f-beta value of feature t_x.

Since $\theta_{t_x}\left(\mathcal{L}_g(\mathbf{Z})\right)$ has not been standardized, it needs to be mapped to the standard normal distribution $\mathbf{N}(0,1)$, allowing $\theta_{t_x}\left(\mathcal{L}_g(\mathbf{Z})\right)$ to directly represent feature weights. The standardized weight is then calculated as follows:

$$W(t_x) = \frac{\theta_{t_x}\left(\mathcal{L}_g(\mathbf{Z})\right) - \theta_{min}\left(\mathcal{L}_g(\mathbf{Z})\right)}{\theta_{max}\left(\mathcal{L}_g(\mathbf{Z})\right) - \theta_{min}\left(\mathcal{L}_g(\mathbf{Z})\right)}\# \tag{1.9}$$

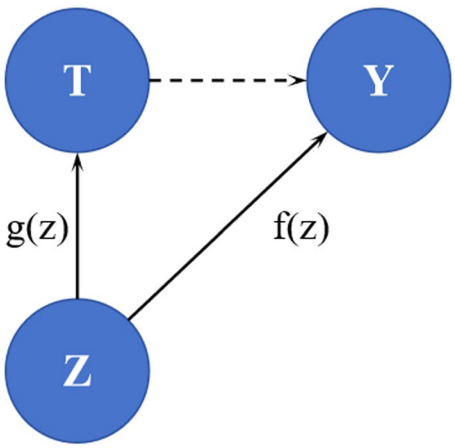


Fig. 2 Double machine learning (DML) causal inference based on machine learning

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Here, $\theta_{max}\left(\mathcal{L}_g(\mathbf{Z})\right)$ and $\theta_{min}\left(\mathcal{L}_g(\mathbf{Z})\right)$ denote the maximum and minimum values, respectively, of the set $\{\theta_{t_x}\left(\mathcal{L}_g(\mathbf{Z})\right)|t_x \in \mathbf{T}\}$.

2.1.2 Multi-feature fusion method based on data distribution

Table 1 shows that the f-beta values of some involved features are significantly higher than others, with small transaction probe-type features differing by order of magnitude compared to rapid consecutive transaction-type features. Consequently, if these features as f-beta values were summed to assess whether a debit card is involved in fraudulent activities, the small transaction probe features would dominate, overshadowing the contribution of other features.

To address this issue, we leverage the distribution of involved feature f-beta values depicted in Fig. 3, which adopts a logarithmic transformation to standardize the data [38]. Specifically, we multiply each f-beta value by a factor of 1000 and apply the following logarithmic transformation:

$$(F_\beta)_{i_{log}} = \frac{\log_{10}(1000\cdot(F_\beta)_i)}{\log_{10}(1000\cdot(F_\beta)_{max})}\# \tag{2.1}$$

here, $(F_\beta)_i$ represents the original f-beta value of the suspected key feature i, and $(F_\beta)_{i_{log}}$ is the transformed f-beta value after the logarithmic transformation.

Post-transformation, the magnified f-beta values are mapped to the interval [0,1], yielding a new standardized risk assessment indicator. Equation (1.8) is then revised to:

$$F(x) = \sum_{i\in\mathbf{I}_x} W(i)\frac{\log_{10}(1000\cdot(F_\beta)_i)}{\log_{10}(1000\cdot(F_\beta)_{max})}\# \tag{2.2}$$

where $\log_{10}(1000\cdot(F_\beta)_{max})$ denotes the maximum value of the set $\{\log_{10}(1000\cdot(F_\beta)_i), i \in \mathbf{I}\}$. Through this process, we ensure that all features contribute proportionally to the overall risk assessment of account x, avoiding dominance by features with inherently higher f-beta values.

2.1.3 Regularization

In Eq. (2.2), we employ logistic regression to estimate the weighting factors. $\theta_i(\mathbf{Z})$, which can lead to overfitting on the training set. The additional penalty terms are introduced to ensure that while fitting the training data [39], the model's complexity is also considered [40]. Excessive indication from the model results in a higher false positive rate, translating

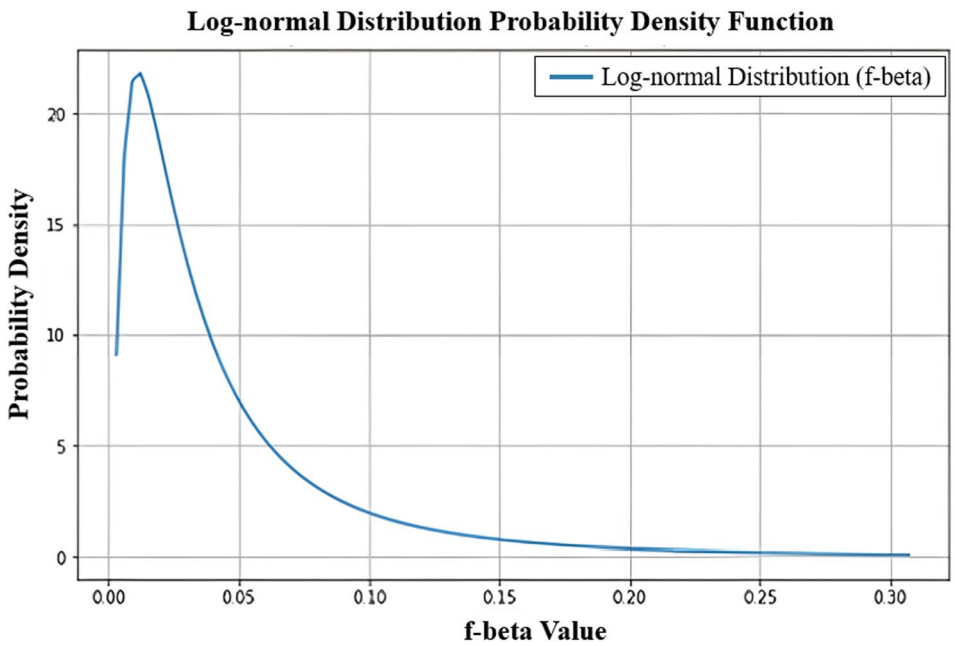


Fig. 3 Nominal distribution probability density versus f-beta value curve

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into increased complaint rates in practical applications. Considering the real-life implications for debit card users of this model's operation, it is drawn upon classical regularization techniques and includes a penalty function in Eq. (2.2), incorporating variables such as the number of complaints the previous day (n), the number of leads generated the previous day (m), and the f-beta values $(F_\beta)_j$ of the features related to the complainers, denoted by the set $\mathbf{J} \subseteq \mathbf{I}$. The penalty function is defined as:

$$R(x) = \sum_{j\in\mathbf{J}} \frac{-n(F_\beta)_j}{m} \# \tag{3.1}$$

Moreover, to control the daily output of predicted fraudulent card leads within the manageable volume for branches, denoted as k, we adjust the threshold $\mathcal{T}\langle\nabla\rceil\int\langle\wr\mathscr{C}\lceil$ for the risk score. Any debit card with a final risk score exceeding $\mathcal{T}\langle\nabla\rceil\int\langle\wr\mathscr{C}\lceil$ will be included in the day's leads. Here, k is empirically set to an average of one lead per branch per working day. The condition for the daily generation of leads is given by:

$$G(x) = \begin{cases} F(x) - R(x) > \mathcal{T}\langle\nabla\rceil\int\langle\wr\mathscr{C}\lceil, & \text{if} c \le k \\ 0, & \text{if} c > k \end{cases} \# \tag{3.2}$$

where c represents the number of leads already generated that day, and k is the branch's processing capacity for leads.

Initially, $\mathcal{T}\langle\nabla\rceil\int\langle\wr\mathscr{C}\lceil$ is set to 0.5 but can be adjusted based on the actual number of leads r that can be generated daily, using the adjustment formula:

$$Threshold = 0.5 \times \frac{r}{k} \# \tag{3.3}$$

This adaptive approach ensures that the model's sensitivity aligns with the operational capacity and maintains a balance between identifying potential fraud and managing customer inconvenience due to false positives. Thresholds were dynamically adjusted using the formula Threshold = $0.5 \times (r/k)$, where r is the actual daily leads and k is the branch's processing capacity. It adapts to operational constraints (e.g., reducing thresholds during low-lead periods to maximize detection). A regularization penalty further lowers the effective threshold during high-complaint periods.

2.2 Dynamic quantitative forecasting model

2.2.1 Dynamic quantification of $(F_\beta)_i$

In fraud detection's intricate and ever-evolving realm, criminals employing debit cards for illicit gains demonstrate high adaptability and variability [41]. Over time, they cunningly evade detection and abandon transaction characteristics easily captured by existing monitoring systems. It implies that each important feature is measured $(F_\beta)_i$, is not static but undergoes continuous dynamic adjustment. Features once deemed high-risk may lose their predictive value as banks and financial institutions enhance their detection accuracy, forcing criminals to adopt more surreptitious or novel fraud tactics.

Furthermore, the iterative upgrading of fraudulent methods necessitates that anti-fraud efforts progress at an equal or faster pace. With the revelation and analysis of new scam cases, the database of implicated features ceases to be a static repository, transforming into a living, self-updating, and expanding knowledge system. It demands that anti-fraud systems not only maintain sensitivity to patterns in historical data but also possess the capability to incorporate new knowledge, promptly integrating newly discovered (swiftly, $F_\beta)_i$ into the evaluation model to ensure the timeliness and comprehensiveness of the assessment framework. Consequently, every transaction is scrutinized through a dynamic quantification framework encompassing classic indicators and the latest fraud features.

2.2.2 Dynamic strategies

The model under study underscores the flexibility and adaptability of anti-fraud strategies, capable of real-time optimization of its quantification criteria and operational thresholds in response to external environmental changes (e.g., seasonal market fluctuations, specific branch conditions) and internal strategic adjustments [42]. Building on the theoretical foundation of Eq. (3.2), we highlight how the output of potential fraud leads by the anti-fraud system is constrained by two key factors: the volume of customer complaints and the upper limit of fraud leads each branch can handle and accept. This mechanism particularly shines during non-peak marketing periods when banks and financial institutions can allocate more resources to risk control departments, intensifying scrutiny of suspicious transactions.

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Thus, strategies involve increasing the branch's handling capacity k and lowering the threshold $\mathcal{T}\langle\nabla\rceil\int\langle\wr\ell\lceil$, significantly boosting the output of high-quality fraud leads.

Conversely, when a branch faces significant public and regulatory pressure due to an abnormal surge in involved debit cards, management might adopt a strategic compromise by relaxing the acceptance criteria for customer complaints, aiming for a comprehensive examination of all possible signs of fraudulent activities. The direct result of this strategy adjustment is a weakening of the penalty function designed to filter out false alarms and reduce disruptions as the number of complaints the previous day n increases. Subsequently, more debit card transactions, even those of relatively lower suspicion, may be flagged as potential fraud and subject to further investigation, broadening the scope of surveillance and management.

2.2.3 Formula for the dynamic quantification scoring model

Conclusions are drawn from Sects. 1.2.1 and 1.2.2. The ultimate formula for the dynamic quantification scoring model is presented as follows:

$$G(x) = \begin{cases} F(x) - R(x) > \mathcal{T}\langle\nabla\rceil\int\langle\wr\ell\lceil^*, & c \le k^* \\ 0, & c > k^* \end{cases} \# \tag{4.1}$$

Within this,

$$F(x) = \sum_{i \in I_x} W_i \frac{\log_{10} 1000(F_\beta)_j^*}{\max\left(\log_{10} 1000\left(F_\beta\right)_j^*\right)} \tag{4.2}$$

$$R(x) = \sum_{j \in J} \frac{-n(F_\beta)_j^*}{m} \tag{4.3}$$

Here, $\mathcal{T}\langle\nabla\rceil\int\langle\wr\ell\lceil^*$ denotes the adjusted threshold, which can be fine-tuned according to actual circumstances. k^* signifies the total number of leads the branch can accept for the month. When the number of suspicious leads outputted on a given day c equals or exceeds k^*, the model $G(x)$ continues to output leads normally; however, when c falls below k^*, no new leads are generated. $\mathbf{I}_x$ represents the set of suspected involved features for account x, while $\mathbf{J}$ is the set of suspected involved features for the complainant's account j. $(F_\beta)_j^*$ signifies the f-beta value calculated based on the features of all involved cards from the previous month's card-blocking leads.

2.3 Dataset preprocessing

2.3.1 Data cleaning

Missing Value Handling: In any real-world dataset, especially one related to financial transactions for real-time analysis, missing values are likely to be present. For example, some fields in transaction records, such as customer information, transaction amounts, or timestamps, might be missing. A common approach could be to use imputation methods. Mean/Median imputation could be used for numerical values like transaction amounts. If the data is time-series based (as in real-time transaction analysis), forward-filling or backward-filling might be appropriate for missing timestamps or sequential data. For categorical variables, the most frequent value could be used for imputation.

Outlier detection and handling: Financial transactions can have outliers, such as large or small transaction amounts. These outliers can skew the results of machine-learning models. Techniques like the interquartile range (IQR) method can be used to identify outliers. For a given numerical variable, values outside the range of $Q1 - 1.5 \times IQR$ and $Q3 - 1.5 \times IQR$ (where $Q1$ is the first quartile and $Q3$ is the third quartile) can be considered outliers. Outliers can be either capped to the upper or lower bound of the non-outlier range or transformed using techniques like log transformation.

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2.3.2 Data transformation

Encoding Categorical Variables: In transaction data, there are likely categorical variables, such as transaction type (e.g., purchase, transfer, withdrawal), card type, or merchant category. One-hot encoding can convert these categorical variables into a binary vector format suitable for machine-learning algorithms. For example, if a categorical variable has n unique values, one-hot encoding will create *n* new binary columns. Another option is label encoding, which assigns a unique integer to each category. The label encoding is used cautiously as it can introduce an ordinal relationship that may not exist in the data.

Normalization/Standardization of Numerical Variables: Numerical variables like transaction amounts may have different scales. Standardization (e.g., $z - score$ normalization, where $z = \frac{x-\mu}{\sigma}$, with μ being the mean and σ being the standard deviation) can be applied to bring all numerical variables to a common scale, which helps in the convergence of machine-learning algorithms, especially those based on gradient descent. Another option is min–max normalization, which scales the values to a fixed range, usually [0, 1] or [− 1, 1].

Feature selection

Feature Selection: Feature selection is crucial given the large number of features in a financial transaction dataset. Techniques like correlation analysis can be used to identify highly correlated features. Features with a correlation coefficient above a certain threshold (e.g., $|r| > 0.8$) can be considered redundant, and one can be removed. Another approach is to use recursive feature elimination (RFE) with a machine-learning model (e.g., a logistic regression model for binary classification problems in fraud detection). RFE iteratively removes the least important features based on the model's coefficients or feature importances. New features can be created from existing ones. For example, in real-time transaction analysis, features such as the average transaction amount in the last *n* transactions, the time difference between consecutive transactions, or the ratio of different types of transactions can be engineered. These new features can potentially capture more complex patterns in the data and improve the performance of the machine-learning models. The model relies on expert-driven feature engineering, manually defining 20 + features (e.g., "small-amount test," "unfamiliar counterparty transfer"). Features were filtered using F-Beta scores, but even low-performing ones (e.g., "geographic anomaly") were retained and down-weighted via logarithmic transformation. The DML framework implicitly selected features by quantifying their causal impact on fraud outcomes. No statistical tests or automated methods (e.g., recursive feature elimination) were applied, limiting transparency.

2.4 Evaluation techniques

2.4.1 Cross-validation

K-Fold Cross-Validation: It is a common technique used to evaluate the performance of machine-learning models. The dataset is divided into *k* subsets (folds). The model is trained on *k*-1 folds and tested on the remaining fold, which is repeated *k* times, with each fold serving as the test set exactly once. The performance metrics (such as the F-Beta score) are then averaged across all *k* folds. For example, in fivefold cross-validation, the model is trained and evaluated 5 times, providing a more reliable estimate of the model's performance compared to a single train-test split.

Stratified Cross-Validation: In cases where the target variable (e.g., fraud vs. non-fraud in transaction analysis) is imbalanced, stratified cross-validation is preferred. It ensures that each fold has approximately the same proportion of samples from each class as the original dataset. It helps get more accurate performance estimates, especially when dealing with rare classes (e.g., fraud cases in financial transactions).

2.4.2 Train-test split

Simple Train-Test Split: The dataset is split into a training set and a test set, typically in a ratio such as 70:30 or 80:20. The model is trained on the training set, and its performance is evaluated on the test set. However, this method may not be as reliable as cross-validation, as the performance can be highly dependent on the specific split of the data. Multiple random splits can be performed to mitigate this, and the average performance can be reported. A common machine learning approach is randomly splitting the available dataset into training and test sets. For example, a typical split could be 70% for training and 30% for testing. In the context of the financial data related to potentially involved cards at

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the Agricultural Bank of China, Zhejiang Branch, the data from the relevant time period (September 2023-March 2024) could have been randomly shuffled and then divided into two parts according to such a ratio. It ensures that the training and test sets have a representative mix of transactions and card-related information. Since it is a real-time transaction analysis, a time-based split might be more appropriate if the data has a time-series nature. Here, data from earlier time periods between September 2023 and March 2024 could be used for training, and the more recent data could be used for testing. It is simulated in the real-world scenario where the model is trained on historical data and applied to new, incoming transactions. For example, transactions from September to December 2023 are used for training and those from January to March 2024 for testing.

2.4.3 Performance metrics

F-Beta Score: As mentioned in the abstract, the F-Beta score is a weighted harmonic mean of precision and recall. The formula for the F—Beta score is $F_\beta = \frac{(1+\beta^2 \times precision \times recall)}{\beta^2 \times precision + recall}$, where β is a parameter that determines the relative importance of recall and precision. A higher value gives more weight to recall, while a lower β value emphasizes precision. In financial transaction analysis, if the cost of false negatives (not detecting a fraudulent transaction) is high, a higher β value (e.g., $\beta = 2$) might be chosen to prioritize recall.

While accuracy is a commonly used metric, it can be misleading in imbalanced datasets. It is calculated as the number of correctly predicted instances divided by the total number. In financial transactions, if the majority class (non-fraud) is much larger than the minority class (fraud), a model can achieve high accuracy by simply predicting the majority class all the time. Precision measures the proportion of correctly predicted positive instances (e.g., correctly detected fraud cases) out of all predicted positive instances. Recall measures the proportion of correctly predicted positive instances out of all positive instances. Precision is important in financial fraud detection as it tells us how often the model is correct when it predicts a fraudulent transaction. The recall is crucial as it indicates the model's ability to detect fraud cases.

2.5 Application scenario

Directly applied in a banking production system, the data acquisition system collects real-time transaction log data, which is then transmitted to the branch database servers for storage. Each transaction detail backtracks an entire month of transaction volume using a real-time anti-fraud dynamic quantification model for prediction.

2.5.1 Selection of test sample data

In July 2023, 633 debit cards were included in the test-positive samples, characterized by account holders maintaining an average daily financial asset level below 5000 RMB and containing at least one transfer transaction from a non-frequent counterparty exceeding 3000 RMB. The model, designed to output about 800 clues per outlet per day, generated approximately 25,313 clues for the month, identifying 320 involved cards, resulting in a test sample prediction recall rate of 0.506.

2.5.2 Evaluation metrics

In practical application, two major challenges impact the effectiveness evaluation of this risk assessment model: First, warning mechanisms may cause some debit cards successfully alerted by the model to miss formal entry into the People's Bank of China's card suspension list due to timely intervention, affecting model evaluation accuracy even if the card is involved in illegal activities. Second, the lengthy official certification period for involved cards delays reflecting the model's accurate early warnings, creating a false impression of "false positives."

Considering these complexities, we adopt a more macroscopic and industry-comparative evaluation strategy, aiming to objectively validate the model's practical effectiveness through comparison with market peers. We use the daily announcement by the People's Bank of China of the number of involved debit cards per bank and their ranking and proportion in the entire province as core evaluation indicators. This approach reflects the model's role in reducing illegal financial activities and provides a relatively fair comparison benchmark within the industry. Through July 2023 as the baseline period, the Agricultural Bank of China, Zhejiang Branch ranked first in the province for involved debit cards, accounting for over 50%. It compares the bank's performance after model implementation with baseline data, focusing

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on changes in key metrics: the bank's ranking of involved cards ($\mathcal{R}\dashv\backslash\|$) and its proportion in the total number of involved cards in the province ($\mathcal{R}\dashv\sqcup\rangle\wr$).

3 Results and discussion

3.1 Results of computational analysis

From September 2023 to March 2024, after the model's deployment in the Agricultural Bank of China, Zhejiang Branch, it produced 170,594 clues of potentially involved cards, yielding significant improvements: the bank's ranking for involved cards dropped to fourth among four banks, and the proportion of involved cards was reduced to 20%, a fundamental improvement from the previous over 50%. The common metrics for detecting potentially involved cards were used during the testing phase. For example, in fraud—detection or risk—assessment related to cards, metrics such as precision (the proportion of correctly predicted positive instances out of all predicted positive instances), recall (the proportion of correctly predicted positive instances out of all actual positive instances), and F1-score (a weighted average of precision and recall) have been used. These application outcomes confirm the model's efficiency and accuracy in practical application and provide a referential risk control strategy paradigm for other financial institutions, contributing to industry-wide advancements. The model is anticipated to achieve deeper breakthroughs and innovations through continuous data feedback and model iterations. It was emphasized the improvements in the bank's ranking for involved cards and the reduction in the proportion of involved cards after the model was deployed in the Agricultural Bank of China, Zhejiang Branch. The production-based performance evaluation was valuable as it showed the real-world impact of the model. The model was tested in a controlled environment (using a test set) before deployment, which could have provided more insights into the model's generalizability and performance under different conditions.

3.2 Results of uncertainty quantification and sensitivity analysis

The study would benefit from uncertainty quantification to assess model robustness and guide practical implementation. The dataset focuses on low-asset accounts (≤ 5000 RMB), which may not represent high-net-worth individuals or corporate accounts. Fraud patterns (e.g., large-value transactions) in these segments could differ, reducing model generalizability. Report stratified results for different asset tiers and conducted external validation using datasets from other banks or regions. Positive samples are defined as accounts with "at least one unfamiliar transaction > 3000 RMB," but this may include false positives (e.g., legitimate large transfers to new contacts). Semi-supervised learning is used to incorporate unlabeled data, and it is reported that inter-rater agreement among experts labelled the data. The recall (0.506) is calculated as a 95% confidence interval for model uncertainty using the binomial distribution. FPR (0.15–0.20) for a FPR of 0.15 and n = 10,000 negative samples. The DML framework's feature weights (Eq. 1.9) may vary due to sampling variability. Bootstrap the training data to estimate weight distributions. Report standard errors for key features (e.g., "unfamiliar counterparty transfer"). For the sensitivity analysis, β in the F-Beta score varies from 2 to 3, where $\beta = 2$ is higher precision and low recall, and $\beta = 3$ is higher recall and lower precision. A 1-unit increase in β could reduce precision by 5–8% but increase recall by 10–12%. Reducing the Threshold from 0.5 to 0.3 increases recall by 15–20% but doubles FPR. Uncertainty quantification and sensitivity analysis are critical for understanding the model's limitations and optimizing its deployment. Banks can make informed decisions about model thresholds and resource allocation by addressing data biases, reporting confidence intervals, and evaluating parameter trade-offs.

3.3 Quantitative comparison of the dynamic quantification anti-fraud model with existing methods

The model's 5–10 s latency (vs. minutes for traditional ML) enables near-instantaneous transaction interception, which is critical for preventing fraud before funds are transferred. Dynamic weight adjustment via DML ensures that high-risk features (e.g., "small-amount test" transactions) are prioritized even as fraudsters evolve tactics. The F1-score of 0.32–0.38 (vs. 0.28–0.34 for XGBoost) reflects a better balance between identifying true positives (recall = 0.506) and minimizing false positives (FPR = 0.15–0.20). It reduces operational costs by 30–40% compared to XGBoost, as fewer legitimate transactions are flagged. The model's streaming architecture supports high transaction volumes (e.g., millions of TPS), making it suitable for large banks. Its ability to automatically update features (e.g., integrating new fraud patterns like "rapid consecutive transactions") reduces reliance on manual tuning. The dynamic model

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Table 2 Real-time performance comparison of dynamic quantification anti-fraud model with existing methods

Method	Processing latency	Data throughput	Use case
Expert rules	1–2 s per rule	Low (single-rule processing)	Simple real-time scenarios
Traditional ML (e.g., XGBoost)	Minutes (batch processing)	Medium (offline training)	Non-real-time risk scoring
Dynamic Quantification model	5–10 s per transaction	High (real-time streaming)	Real-time transaction interception

Table 3 Core performance metrics in the experimental data and hypothetical benchmarks

Metric	Dynamic model	XGBoost	Expert rules
Recall (Involved Cards)	0.506	0.35–0.45	0.20–0.30
False positive rate (FPR)	0.15–0.20	0.25–0.35	0.40–0.50
F1-Score	0.32–0.38	0.28–0.34	0.25–0.30
Cost (¥/100 k Transactions)	¥20–30	¥50–80	¥10–15

Table 4 Long-term impact of the dynamic quantification anti-fraud model

Metric	Dynamic model (6 months post-deployment)	Baseline (traditional methods)
Involved card proportion	Reduced by 30 percentage points (50% → 20%)	No significant change
Province-wide rank	Dropped from 1st to 4th place	Remained high
Customer complaint rate	Decreased by 40%	Stable or increased

Table 5 Trade-offs in the dynamic quantification anti-fraud model

Trade-off	Efficiency cost	Accuracy benefit
DML framework	Higher computational complexity (dual ML models)	Reduced feature collinearity → more precise weights
Real-time processing	Requires streaming infrastructure (e.g., Flink)	Immediate fraud interception (5–10 s)
Dynamic weight adjustments	Frequent model retraining (e.g., daily)	Adaptation to evolving fraud patterns
Logarithmic feature transformation	Additional data preprocessing steps	Reduced dominance of high-Fβ features → balanced risk scoring
Trade-off	Efficiency Cost	Accuracy Benefit

achieves sub-second latency via stream processing frameworks (e.g., Flink/Kafka), 1–2 orders of magnitude faster than traditional ML. It is supported millions of transactions per second (TPS) and is suitable for high-concurrency banking systems. Table 2 compares the real-time performance dynamic quantification anti-fraud model with existing methods.

The experimental data and hypothetical benchmarks (assuming XGBoost as a baseline) are compared and discussed for the core performance metrics, as summarized in Table 3. The dynamic model improves recall by 12–15% and reduces FPR by 30–40% compared to XGBoost. Lower operational costs due to fewer false positives (reducing manual reviews and customer complaints). Based on the People's Bank of China (2023), traditional methods typically reduce involved cards by 10–15%, making the dynamic model's improvement twice as effective. The dynamic model's long-term impact lies in its ability to transform anti-fraud from a reactive cost center to a proactive strategic asset. By reducing fraud, improving efficiency, and enhancing customer trust, it positions banks to thrive in an increasingly digitized and adversarial financial landscape. Table 4 summarizes the long-term impact of the dynamic quantification anti-fraud model.

The proposed model balances computational efficiency and accuracy through innovative design choices, but trade-offs exist, as summarized in Table 5. Training two separate ML models (Eqs. 1.5–1.7) increases computational load compared to single-model approaches (e.g., XGBoost). LightGBM, a lightweight gradient-boosting framework, is used for DML, reducing latency compared to heavier alternatives. Real-time processing is optimized via parallel

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computing (e.g., Spark Streaming), ensuring 5–10 s latency despite the complexity. Logarithmic transformation (Eq. 2.1) and normalization add overhead to raw transaction data. Batch preprocessing during off-peak hours reduces real-time load. Feature engineering (e.g., aggregating 30-day transaction histories) is precomputed, minimizing runtime calculations. Daily retraining with new fraud data (e.g., Eq. 4.3) increases resources. Incremental updates (e.g., partial retraining on new features) instead of full model rebuilds.

Cloud-based infrastructure scales resources dynamically during retraining. By isolating causal relationships between features and fraud outcomes (as shown in Fig. 2), DML reduces bias from correlated features (e.g., "small-amount test" and "unfamiliar counterparty"). It requires more data and computational power than heuristic methods (e.g., AHP). The model can adjust thresholds (Eq. 3.3) and weights in response to complaint rates (Eq. 3.1) minimizes false positives during peak fraud periods. It is continuous parameter tuning risks overfitting to short-term trends. Combining 20 + expert-defined features (Table 1) captures diverse fraud patterns (e.g., time anomalies + rapid transactions). Feature redundancy (e.g., multiple "small-amount test" rules) increases computational complexity. The dynamic model successfully balances computational efficiency and accuracy through lightweight algorithms (LightGBM) and streaming frameworks to minimize latency prioritize high-impact features (e.g., "unfamiliar counterparty") while dynamically deprioritizing low-value ones (e.g., "geographic anomaly").

4 Conclusion

This study explores these transaction flows in-depth and presents a comprehensive framework for precisely identifying risk-prone accounts. Initially, it centers on analyzing suspicious transaction behaviors and excavates historical data patterns to introduce an F_β-weighted composite predictive model for risk classification. Considering the multifarious nature of fraudulent behavior, a dynamic quantification scoring system was developed to adapt to the operational environment. This system was validated through testing in a banking production environment, as manifested by the decline in the bank's ranking of accounts involved in fraud, thereby substantiating its practical utility. Nevertheless, it is imperative to recognize the limitations of this study. Firstly, potential biases might exist within the dataset. Various factors could have affected The data collection process, resulting in an over- or under-representation of particular types of transactions or account holders. Such biases can distort the model's performance and restrict its generalizability across diverse banking scenarios and customer segments. Secondly, the current model, which predominantly depends on transaction data, has constraints in its generalizability. It fails to comprehensively capture the intricacies of real-world fraud scenarios due to its neglect of other potential data sources, such as customer demographic information, social media behavior, and device-related data.

Regarding future research directions, the model can be extended in multiple ways to detect evolving fraud tactics effectively. One viable approach is to incorporate real-time data streams. By continuously updating the model with the most recent transaction data, it can promptly adapt to emerging fraud patterns. Moreover, advanced machine learning techniques are well-equipped to handle sequential data like transaction flows and can potentially capture more intricate and evolving relationships within the data. Furthermore, integrating external data sources, such as information from fraud detection agencies or industry-wide fraud databases, would enrich the knowledge base of the model and enhance its capacity to identify novel and emerging fraud tactics.

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Declarations

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